CS3332 Software Engineering

Midterm Exam

Lecturer: Tuan-Dung CAO
Duration: 100 minutes

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1. (15 pts)

a. What is *software engineering*?
Present your opinion about the necessity and importance of software engineering, especially in modern software development.

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2. (15 pts)

Present **at least three erroneous beliefs (myths)** about software or the process used to build it, and explain the **corresponding truths in reality**.

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3. (25 pts)

a. What is a **software process**?
Compare **two generic software process models** that you know. *(15 pts)*

b. Propose the **most appropriate software process model** that could be used to manage the development of the following systems and explain your reasons. *(10 pts)*

1. A system to control and monitor **nuclear power plants**.
2. A **Learning Management System (LMS)** with integrated video conferencing features from third-party services such as Zoom or MS Teams.
3. A **digital marketing tool** that helps online sellers and small companies boost their business.

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4. (15 pts)

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Explain **in detail two** of the following key tenets of **Extreme Programming (XP)** and describe **why each is important**:

a. Pair Programming
b. Constant Unit Testing
c. Short Iterative Cycles
d. Stories as Requirements
e. The 40 Hour Week
f. Refactoring

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5. (30 pts)

You are part of a **Requirement Engineering (RE) team** developing an **online COVID-19 health declaration application**.

Write the following for the system:

a. At least **three user requirements**.

b. Choose **one** of them and break it down into **system requirements**, then describe those system requirements.

c. Provide **two measurable non-functional requirements**.